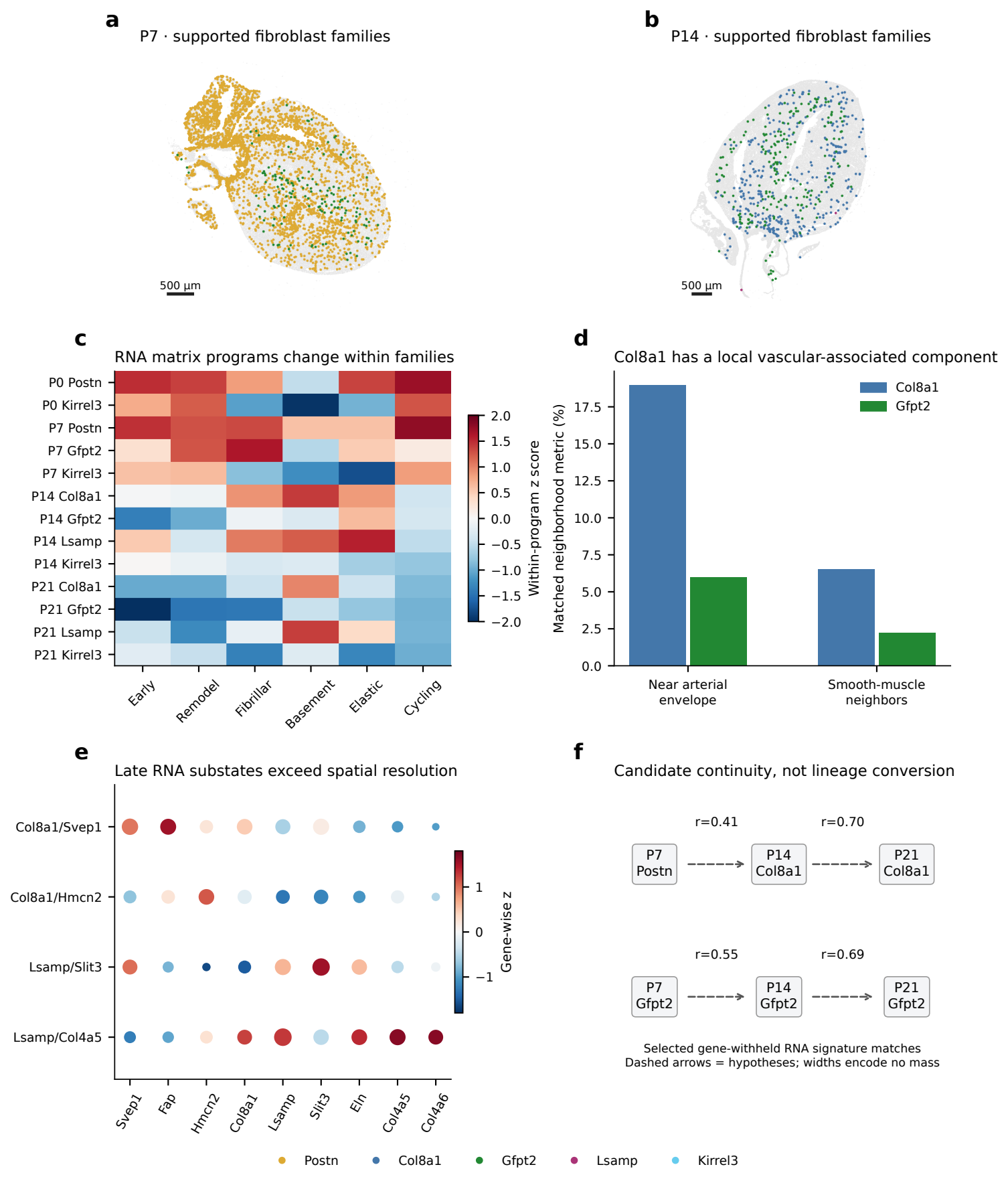

a
P7 · supported fibroblast families
500 µm
b
P14 · supported fibroblast families
500 µm
c
RNA matrix programs change within families
P0 Postn
P0 Kirrel3
P7 Postn
P7 Gfpt2
P7 Kirrel3
P14 Col8a1
P14 Gfpt2
P14 Lsamp
P14 Kirrel3
P21 Col8a1
P21 Gfpt2
P21 Lsamp
P21 Kirrel3
Early
Remodel
Fibrillar
Basement
Elastic
Cycling
Within-program z score
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
d
Col8a1 has a local vascular-associated component
Col8a1
Gfpt2
Matched neighborhood metric (%)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Near arterial envelope
Smooth-muscle neighbors
e
Late RNA substates exceed spatial resolution
Col8a1/Svep1
Col8a1/Hmcn2
Lsamp/Slit3
Lsamp/Col4a5
Svep1
Fap
Hmcn2
Col8a1
Lsamp
Slit3
Eln
Col4a5
Col4a6
Gene-wise z
1
0
−1
f
Candidate continuity, not lineage conversion
r=0.41
r=0.70
P7 Postn
P14 Col8a1
P21 Col8a1
r=0.55
r=0.69
P7 Gfpt2
P14 Gfpt2
P21 Gfpt2
Selected gene-withheld RNA signature matches
Dashed arrows = hypotheses; widths encode no mass
Postn
Col8a1
Gfpt2
Lsamp
Kirrel3